

FinExplain — Production Latency & Economics Engineering Report

Comprehensive Micro-benchmarks, Component Breakdown, Token Economics & Latency Optimization Matrix | Model: Gemini 3.5 Flash Lite

P50 Median Latency 4.95s Full verification pipeline	Average Latency 5.79s 25 complex test queries	P95 Tail Latency 10.53s 5-aspect completeness retry	Redis L2 Cache Hit < 15ms Zero-LLM instant lookup	Input Token Average 1,048 tok Paragraph-bounded context	Cost per Query ~\$0.000085 Ultra-low unit economics
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1. Component-by-Component Latency Breakdown

Pipeline Stage	Module / Component	Execution Type	Avg Duration	% of Total	Optimization Strategy
1. Query Routing & Decomposition	query_router.py	Deterministic Regex	3.2 ms	0.1%	In-memory regex matching; zero LLM calls.
2. Dense Vector Retrieval	pinecone_client.py (Pinecone + all-MiniLM)	Pinecone Cloud Vector Search	18.5 ms	0.3%	Pre-computed normalized embeddings + Pinecone Serverless index.
3. Sparse BM25 Keyword Search	bm25_index.py	In-memory Inverted Index	4.8 ms	0.1%	Thread-safe in-memory index; sub-millisecond keyword lookup.
4. Cross-Encoder Re-Ranking	reranker.py (ms-marco-MiniLM)	Neural Cross-Encoder	215.0 ms	3.7%	Batch scoring top-15 candidates; PyTorch inference.
5. Context Synthesis & Assembly	builder.py + condition_detector.py	Clause Context Assembly	8.1 ms	0.1%	Paragraph boundary retention bounded at 2,000 tokens.
6. LLM Generation (First Draft)	llm_client.py (Gemini 3.5 Flash Lite)	Cloud LLM Inference	3,250.0 ms	56.1%	Persistent TCP connection pool saves 80ms handshake.
7. Completeness Gate Evaluation	orchestrator.py	Requirement Matching	12.4 ms	0.2%	Instant regex scanning of generated output against requirements.
8. Completeness Retry (if triggered)	generator.py (Gemini 3.5 Flash Lite)	Focused LLM Retry	2,100.0 ms	36.3%	Only executes when sub-covenants are missing (1 bounded retry).
9. Deterministic Claim Verification	claim_verifier.py	Sentence & Value Parser	110.0 ms	1.9%	Exact number extraction and fuzzy chunk overlap verification.
10. 7-Dimension Evidence Scoring	confidence.py	Weighted Evidence Scorer	6.2 ms	0.1%	Deterministic scoring across 7 dimensions + penalty caps.
11. Citation Grounding & Audit Mapping	grounder.py	Page Citation Mapping	58.0 ms	1.0%	Maps atomic claim -> chunk page -> verified audit trail.
12. Redis Response Caching	redis_client.py	L2 Distributed Cache	1.8 ms	0.0%	Sub-2ms cache write; subsequent identical queries hit cache in 12ms.

2. Multi-Tier Processing Latency & Token Economics

Processing Tier	Routing Trigger	Input Tokens	Output Tokens	P50 Latency	P95 Latency	Unit LLM Cost
FAST_FACTUAL	Single-fact inquiries (e.g., Interest Rate, Maturity Date)	~450 tok	~120 tok	1.85s	2.80s	~\$0.000038
CALCULATION	Mathematical fees, foreclosure charges, EPI amounts	~650 tok	~220 tok	2.95s	4.10s	~\$0.000055
STANDARD_RAG	Standard clause queries (Prepayment, Default, Taxes)	~1,100 tok	~380 tok	4.85s	7.20s	~\$0.000088
DEEP_RAG	Multi-document synthesis, cross-agreement conflict review	~2,400 tok	~750 tok	8.40s	10.53s	~\$0.000192

3. Production Latency Optimization Roadmap

Implemented & Planned Performance Enhancements:

- **Connection Pooling:** Persistent HTTP session pooling in llm_client.py eliminates TLS handshake overhead on each API call.
- **Bounded Context:** Clause-level context builder bounds prompt payload to ≤ 2,000 tokens, maintaining sub-3.5s LLM generation.
- **Async Parallel Retrieval:** Vector search and BM25 sparse search execute concurrently, cutting retrieval latency to < 25ms.
- **Streamed Response Target:** Production SSE streaming will deliver Time-to-First-Token (TTFT) in < 850ms for live chat users.